

The Addiction Game: Prevention and Unstable Preferences

Levi Lowe

Department of Philosophy, University of Utah

Statements and Declarations

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Correspondence concerning this article should be addressed to Levi Lowe, University of Utah, 201 Presidents' Cir, Salt Lake City, UT 84112, United States.

Email: levi.lowe@utah.edu

ORCID iD: 0009-0001-0095-9399

Abstract

Addictive behavior is often considered irrational. The argument goes: it does not make sense that one would continuously use something, like drugs or alcohol, that runs orthogonal to her preferences or otherwise frustrates her interests. It is this self-induced contradiction of preferences that philosophers and clinicians describe as “irrational.” While there is opposing literature that posits a theory of rational addiction, the word “rational” here always refers to agents with forward-looking and stable preferences. Using a simple game theoretic setup, I show how even in cases involving *unstable* preferences, rational choices do not prevent addiction.

Keywords: rational decision making, game theory, addiction, preferences, prevention

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1. Introduction

In recent addiction literature, addiction has been described as involving some level of irrationality (Holton and Berridge 2013, Pickard and Ahmed 2017, Dembić 2021, Sripada 2022). Here, “irrational” means “preference contradicting.” That is, in the model of addiction I am focusing on, preference misalignment is the key feature of addiction. Yet, those authors do not conclude that addiction prevention is about being rational (i.e. preference-aligned). In this paper, I will explain why that is.

My thesis here importantly differs from that of current rational addiction literature, specifically Becker and Murphy (1988).¹ While Becker and Murphy (1988) do model addiction as a rational process, they also model addiction as a deterministic and cumulative process, where the agent maintains stable preferences over time. Those modeling choices are important because, under those assumptions, abstention is always (at least weakly) rational at the prevention stage. I model addiction as a probabilistic process, where the agent has unstable preferences. In doing so, we find cases wherein abstention is irrational at the prevention stage. The elimination of the strategic dominance of abstention at the prevention stage can explain, in some cases, how addiction can arise from rational choices. Becker and Murphy (1988) explain that an addiction, when present, could be sustained rationally. I am here interested in the prevention stage, when the addiction is not already present.²

¹For more, see: Ainslie (1991), Becker et. al. (1991), Mele (1992), Mele (1996), Becker & Murphy (1998), Levy (2006), and Heather & Segal (2015).

²On top of that, economist Ole Røgeberg (2004) argues that many, if not all, economic “rational addiction” models *fail* due to their adherence to “well-informed and forward looking rational addicts with stable preferences, whose addictions are time-consistent, optimal consumption plans” (2004). I simplify the game theory math, making the argument more accessible to students wishing to understand this piece of the conversation.

Modern philosophy of addiction suggests that a host of measures prevents addiction. Bryan et al. (2010) discuss commitment devices, for example. A commitment device targets certain desires (e.g. one's desire for alcohol) and makes them unable to be acted upon (e.g. taking a pill that induces vomiting when mixed with alcohol). Pickard (2018) suggests a more holistic approach. She posits an ecosystem of services that drugs provide. So we seek to understand the reasons one feels drawn to one's addiction (e.g. pain management, identity, socioeconomic opportunities), and we address those. Flora (2022) writes that addiction is best prevented using several methods, including, but not limited to: community-based cognitive behavioral approaches, youth development programs, peer-to-peer counseling, and narrative therapy. Griffin and Botvin (2010) explain how a three-pronged approach to prevention—school-based, family-based, and community-based—can be effective. Most agree that addiction prevention is neither simple nor formulaic.

As many have pointed out, many people with addiction—either substance-induced or not—are unreliable at addiction risk-assessment. Comorbidity with other economic and social stressors also makes risk assessment more challenging. Perhaps it is for these reasons that one develops an addiction. However, to be charitable to one who suspects that rational choices prevent addiction, imagine someone whose ability to make those calculations is at least fair. In fact we will take a *perfectly rational* (defined as preference-aligned) individual and walk with them toward addiction. The use of a perfectly rational individual here is not to demonstrate how real people actually operate. Again, it is to be as charitable as possible to one who suspects that rational choices prevent addiction.

The irrationality model of addiction claims that a person with addiction acts rationally on the calculations and values in their mind, but that those calculations are both skewed and

unresponsive to negative feedback (Ainslie 1991, Holton and Berridge 2013, Pickard 2018, Sripada 2022). So, the feature of the operating view that allows addiction to be characterized as irrational is that addictive desires remain fixed in the face of preference-contradicting results (Holton and Berridge 2013, Dill and Holton 2014). Because these kinds of desires are rigidly responsive to failures in instrumentation, it might be fair to refer to them as irrational—where “irrational” means preference contradicting. And then, if we characterize addiction as irrational, it may seem like a fair next step to think that remaining rational would at least be tantamount to preventing these kinds of irrational desires from operating. So, why is that not the dominant view? In what follows, I use game theory to show that being rational will not prevent addiction.

I will first introduce a game theoretic model of addiction (Dixit et. al. 2015), where ‘just say no’ is the rollback equilibrium—that is, the optimal choice when all players are seeking their highest payoff. Then, I revise the game to include the possibility of non-addiction dependence. Briefly, a non-addiction dependence is a dependent-style relationship, like one’s dependence on water or medicine, that is not an addiction. I do not mean “dependence” in the sense that Charles O’Brien (2011) talks about it. I mean normal, everyday, cases of relying on something to achieve a goal. It is what Sripada (2022) calls “rational” use. Adding this element to the game shifts the rational choice away from ‘just say no’ to instead further engagement with the potentially addictive substance. Then I further modify the payoffs to show how that affects the rational choice, but does not affect my claim.

2. The Original Addiction Game

I am going to model the possibility of addiction as a sequential move game between one’s Current Self (Player 1) and one’s Future Self (Player 2). Here, I model the self as two agents

because the current self and the future self have different, or unstable, preferences as one develops an addiction. Thus each version of the self has distinct payoff functions. For example, perhaps Jane originally smokes tobacco to enjoy spending time with her friends. However, over time, Jane smokes tobacco to stave off withdrawal symptoms. These two versions of Jane are differentiable across both time and preferences, or payoff functions.

What follows are the rules to Dixit et. al's (2015) addiction game. Each player gets one turn. There are two moves to choose between. Current Self goes first. Current Self can either *engage* with, or *disengage* from, a given potential addiction. If Current Self engages, then it becomes Future Self's turn whose moves are also to either *engage* or *disengage*. If Current Self disengages, the game is over before Future Self gets to move. That is the game. What is Current Self's optimal strategy? To find this answer, we need to know the payoffs to each player for every possible outcome. However, spoiler-alert, according to Dixit et. al. (2015), the optimal strategy is to *just say no*; just say no to drugs and alcohol. Dixit et. al (2015) provide a diagram of this game, showing how each player's payoffs would lead to such a disengaged optimal strategy for Current Self (Figure 1).

Dixit et. el. (2015) begin by explaining the reasons behind the shown payoffs. We can start with Current Self.

Choose the outcome of never smoking at all as the standard of reference, and call its payoff 0. There is no special significance to the number 0 in this context; all that matters for comparing outcomes, and thus for [Current Self's] decision, is whether this payoff is bigger or smaller than the others. Suppose [Current Self] best likes the outcome in which she tries smoking for a while but does not continue. The reason may be that she just likes to have experienced many things firsthand or so that she can more convincingly be able

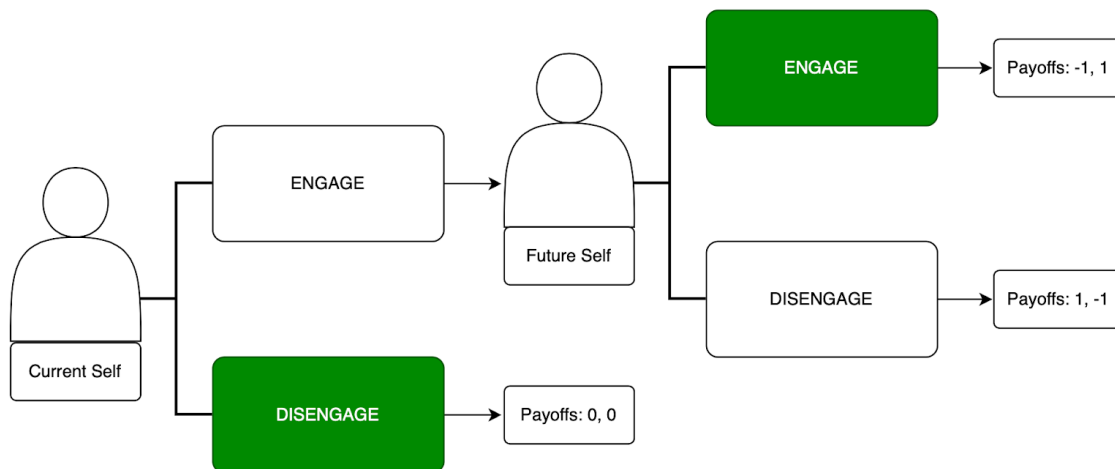


Fig. 1 A sequential move game between Current Self and Future Self. Each player gets one turn, and a choice between two moves: to either engage or disengage. Current Self goes first. If, at the first node, she disengages, the game is over (we end up at the southernmost node). If she instead engages, it is then Future Self's turn, whose options are the same.

Payoffs for each player are written beside each of the three terminal nodes, in pairs (Current Self's payoff is written first). The rollback equilibrium is for Current Self to disengage and Future Self to engage (indicated by green boxes). Since Current Self will disengage, Future Self will not be allowed a turn (app.diagrams.net)

to say "I have been there and know it to be a bad situation" when she tries in the future to dissuade her children from smoking. Give this outcome the payoff 1. The outcome in which she tries smoking and then continues is the worst. (Dixit et. al. 2015)

Now let us go over Future Self's payoffs.

At the initial node, [Current Self] decides whether to try smoking. If her decision is to try, then the addicted [Future Self] comes into being and chooses whether to continue... The payoffs of [Current Self] are as before. But [Future Self] will enjoy continuing to smoke and will suffer terrible withdrawal symptoms if she does not continue. Let [Future Self's] payoff from [Engage] be 1 and that from [Disengage] be -1. (Dixit et al. 2015)

With the payoff structure provided, and accompanying game tree, we can write two expected payoffs for Current Self: one for when she engages, and one for disengagement.

Current Self's expected payoff from engagement = -1

We know Current Self's payoff from engagement will be “-1” because Future Self will choose the outcome that leads to Future Self's highest payoff, namely additional engagement—where Future Self gets a payoff of “1” and Current Self gets a payoff of “-1.” Now we need to compare “-1” to the payoff we expect from Current Self's *disengagement*.

Current Self's expected payoff from disengagement = 0

To find the optimal strategy, we determine which expected payoff is higher and adopt that strategy. Between the two expected payoffs, Current Self's expected payoff from disengagement is greater than her expected payoff from engagement. Therefore, in the rollback equilibrium, Current Self will *disengage*, or *just say no*, because she knows that Future Self will lead them both to an addiction if Current Self engages. On the flipside of that conclusion, we can say that the development of an addiction results from irrational decision making by Current Self. That is one explanation of how addiction develops through irrational choices—and thus where being rational would prevent addiction.

Here is where I revise the game. Of course, the rational decision is to *just say no* if a substance is sure to lead to addiction. The authors say, “If her decision is to try, then the addicted [Future Self] comes into being...” In other words, Dixit et. al's (2015) game tree assumes that

engagement always leads to addiction. They assume that there is a 100% chance of addiction. This assumption does not hold. I will show that, by incorporating the odds that a relation ends up a non-addiction dependence, rather than an addiction, the rollback equilibrium changes; there become many more circumstances in which Current Self's optimal strategy is to *engage* or *say yes*. In terms of game theoretic notions, by “non-addiction dependence,” I am referring to rational engagement (preference aligned choices); and an “addiction” refers to irrational engagement (preference contradicting choices). These sorts of non-addiction dependencies are uncontroversially beneficial, like one's dependency on water, food, shelter, certain medications, or even on friends. While, as Becker and Murphy (1988) point out, these things have the potential to become addictions, regular engagement with them is often part of a balanced lifestyle.

3. Incorporating Non-Addiction Dependence into the Addiction Game

In this new version of the addiction game, I give credence to the possibility that choosing to depend on something may not always lead to an addiction. This means that the Future Self's additional engagement may not always lead to Current Self's worst-case scenario. There is a chance it would lead to both players' best-case scenarios: some non-addiction dependence (hereafter called “dependence”). Dixit et. al. (2015) had left this possibility out of their version of the game.

Now “Chance” enters as the game's third player. Chance gets the first move: either set the players up with what will become a dependence or what will become an addiction. We will call the chances that something ends up as an addiction “ p ”—which will be a number between 0 and 1—and the chances that it ends up a dependence “ $1 - p$ ”. Thus, when p is 0.5, there is a

50%/50% chance of either relation. If $p = 0.6$, there is a 0.4 (40%) chance of dependence. So on and so forth. All this does is incorporate the possibility of non-addiction dependence into Dixit et. al.'s (2015) model.

Additionally, Current Self does not know the result of chance's move—but knows the odds of the relations (e.g. $p = 0.7$). This means that one could say to oneself, “There's a 70% chance that I will become an alcoholic if I start drinking, but I won't know until it's too late.” No amount of further investigation or deliberation can key Current Self into whether the substance (or whatever it is) is going to lead to an addiction. This sort of gamble that Current Self takes conservatively simulates the fact that we cannot see the future, but can sometimes have rough estimates.

There are now two sets of outcomes whose payoffs we need to assign: the set which arises from addiction, and that which arises from dependence. The set of payoffs that arises from addiction is the same as in Dixit et. al.'s (2015) model (Figure 1). The other set of payoffs, which arises from dependence, need to be assigned. Since we search for, and cherish, things to successfully depend on, Current Self avoids disengaging from dependable things, so doing so gives Current Self a payoff of -1. When Future Self chooses to disengage from a dependence, the expected payoffs for each player are the same in both game trees: Current Self gets the short term benefit of its use (+1), and Future Self is expected to experience something like withdrawal—it will not be easy to stop a dependence—so Future Self gets a payoff of -1. If Future Self engages with a dependence, Future Self would get a positive payoff of +1; Current Self would get a payoff of +2 (or at least more than the payoff from a single use because she prefers lasting dependencies over single uses).

With this payoff structure, when $p = 0.5$, we get the following game tree (Figure 2).

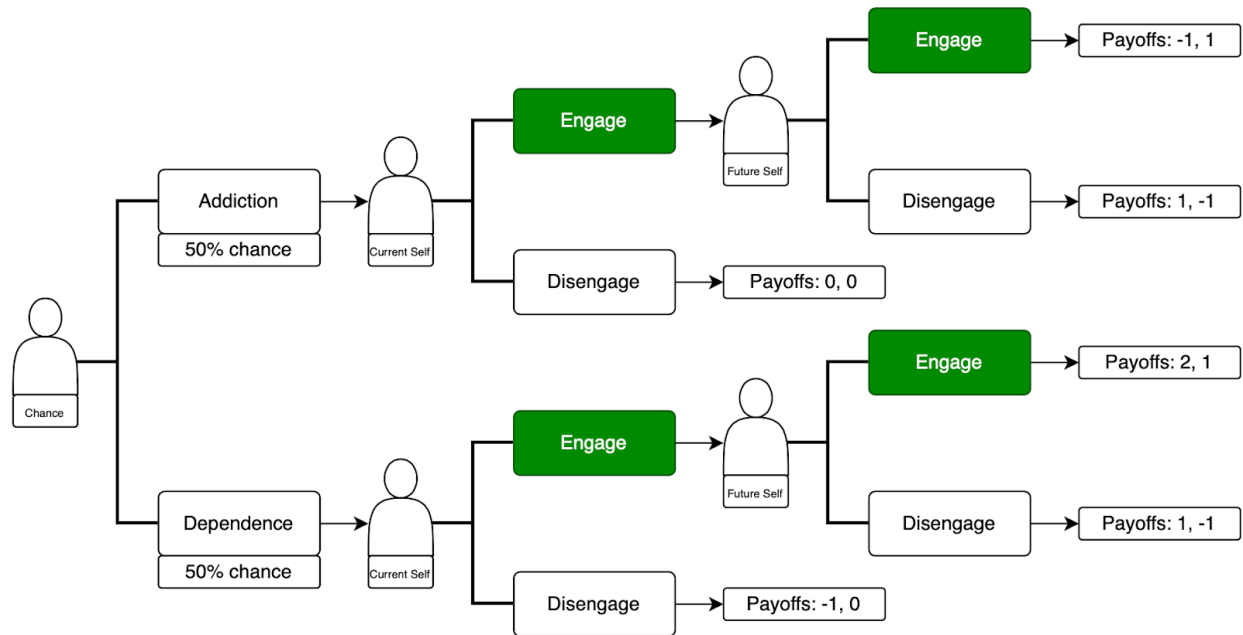


Fig. 2 A sequential move game between Current Self and Future Self. Chance gets the first move. There is a 50% chance of either dependence or addiction. Moving second is Current Self. Current Self does not know the result of Chance's move, but must either engage or disengage. If Current Self disengages, then the game is over. If Current Self engages, it becomes Future Self's turn who does indeed know whether a dependence or addiction has developed. Payoffs for each player are written beside each of the three terminal nodes, in pairs (Current Self's payoff is written first). Future Self's dominant strategy is to engage. Here, the rollback equilibrium is for both players to *engage* (app.diagrams.net)

I am going to explain how to find the rollback equilibrium. Firstly, we need to recognize that Future Self has a dominant strategy: she will always engage. She will engage in dependence and engage in addiction. In this model, the difference between the two chance outcomes matters only for Current Self, who wishes to engage with dependencies and disengage from addictions. This might sound absurd, realistically. In real life, it is not the case that people always keep engaging with something that they engage with once. People often stop using things. Of course, this would be a weakness of the model if the goal of the model were to show how engagement or addiction works in general. However, I am not showing that. I am showing that the mere

inclusion of non-addiction dependence into the addiction game makes it so that abstention is not strategically dominant at addiction's prevention stage. In other words, I could provide even more outcomes from engagement and their respective payoffs. I could consider some "neutral non-dependence" for example (i.e. something that you do not find useful nor very costly). But I believe that this would only complicate the game without a good reason. I would be trying to make the very same point I am making without it, and would only be incorporating it for the sake of a realism that is not germane to my argument. I am showing how merely including non-addiction dependence is enough to make that point. Adding more outcomes would be fuel to that point (namely, again, that rationality may not prevent addiction). So, while Future Self's strategic dominance of engagement is unrealistic, it is an acceptable side-effect of the model's simplicity, and is chosen in pursuit of the model's purpose.

Now, to determine Current Self's optimal strategy, we need to determine Current Self's payoffs from her two options. Since we know Future Self's dominant strategy, we can write out Current Self's expected payoff functions.

When Current Self engages, expected payoff (note: the asterisk means multiply):

$$(chances\ of\ addiction) * (payoff\ of\ addiction) + (chances\ of\ dependence) * (payoff\ of\ dependence)$$

$$\text{When Current Self engages, } expected\ payoff = p(-1) + (1 - p)(2)$$

When Current Self disengages, expected payoff:

$$(chances\ of\ addiction) * (payoff\ of\ addiction's\ disengagement) + (chances\ of\ dependence) * (payoff\ of\ dependence's\ disengagement)$$

When Current Self disengages, *expected payoff* = $p(0) + (1 - p)(-1)$

The rational choice for Current Self will be to engage when the expected payoff from engagement is greater than or equal to the expected payoff from disengagement—in other words, when there is no benefit to switching strategies. That is the rollback equilibrium. Therefore, algebraically speaking, Current Self should engage when:

$$p(-1) + (1 - p)(2) \geq p(0) + (1 - p)(-1)$$

Or, simplified, when

$$p \leq 3/4$$

The only variable in this inequality is “p”—the odds that something will be an addiction. When we solve for “p” given the current payoff structure, Current Self should engage when the odds of an addiction are 75% or lower. We will call “ $p = 0.75$ ” an *inflection point* for this inequality, where optimal strategies change. One important feature we should note is that, when $p = 0.7$, for instance, while it is optimal to engage, one will probably develop an addiction. However, this is the strategy that neither player would do better to switch from because their expected payoff from disengagement is lower. Therefore, given that we benefit from dependencies, the development of addiction may result from rational choices. Of course, this is a very specific case, with specified probabilities and payoffs. Indeed, not every addiction is a result of this strategy. Rather, I have provided a *possibly-so* story, showing that, at the very least, the development of an addiction is not necessarily the result of irrational choices.

There is one inconsistency I should address before moving on. The kinds of preferences that I am talking about when I talk about Future Self are local preferences, while the preferences for Current Self are global preferences. So, Current Self is thinking, *I want to smoke a cigarette, but I do not want to become addicted. So, if I do become addicted, that would contradict my overall goals in life.* That is why, if Future Self does become addicted, then Current Self gets a negative payoff. But where is the negative payoff for the future addict? Addiction is at least in part characterized by a person experiencing a negative payoff to their global preferences. Yet, in my model, the Future Self who chooses to engage with an addiction gets a positive payoff—or at least a better payoff than disengagement at the current stage. We might therefore think that, for the model to be right, the payoff for Future Self who engages with an addiction needs to be negative.

Unfortunately, there is a worse problem with that solution—there would then be no rational reason for Future Self to engage with an addiction. This would be consistent with the addiction literature that says addicts are making decisions that do not align with their global preferences; but a model made this way would inaccurately suggest that the misalignment manifests as irrational decision making. The misalignment of preference that occurs in addiction is explained as either involving the hijacking of dopaminergic (or other motivational) systems, or as some other prediction-based skill-issue on the part of the user—not as knowingly making irrational decisions. Holton and Berridge (2013), Dill and Holton (2014), Pickard (2018), Dembić (2021), and Sripada (2022) have all made the case that addiction could result from someone miscalculating the actual value of a potential reward, rather than one actively and knowingly acting contrary to her preferences. For example, in a case of hyperbolic temporal discounting, the individual's prediction of how valuable, say, a certain drug would change

drastically depending on her temporal proximity to the drug in question (Pickard 2018). This skew in the person's assessment skills could result in them making choices which oppose her actual preferences. If her assessment of the drug during close proximity exaggerates the benefits of use, then it could lead her to voluntarily take it in order to relax, only to find herself experiencing stress.

The word "irrational" describes the desire for the addiction, not the choice to engage with it. As was briefly mentioned in the introduction, this means that, even if one takes a drug and discovers it unpleasurable, that does not necessarily teach her not to use the drug. Holton and Berridge (2013) and Dill and Holton (2014) dive especially deep into exactly how, neurologically, one's desire for something can get divorced from her knowledge of whether she likes that thing. This way, one may crave what she knows she does not like (or refuse to pursue what she knows she would enjoy). These authors posit that one's mesolimbic dopamine systems are responsible for the strength of wanting, but not of liking. They posit that, in addiction, the dopamine system gets hijacked, incorrectly telling the 'liking system' that a given stimulus will be rewarding—"hijacked" because the signal is inaccurately exaggerated. Holton and Berridge (2013) say, "Since these are intrinsic and not instrumental desires, they are not undermined by the belief that consumption of the drugs will not be pleasurable, or that it will be harmful in some other way." Because these intrinsic desires, or what Dill and Holton (2014) call "incentive salience desires," are rigidly responsive to failures in instrumentation, they are referred to as "irrational."

It is for that reason that it would not be consistent with the rational choice model to make the payoff for Future Self, who engages with an addiction, negative; the payoffs in rational choice models are supposed to explain why the person makes a particular decision. So, if Future

Self's engagement with an addiction produced a negative payoff, then it would no longer be explanatory of why Future Self chooses that option. With the positive payoff, I can capture the fact that there is something like a local or short-term goal that Future Self is acting upon in order to engage with an addiction. So, I switch between those two kinds of payoffs: global preferences for Current Self and local preferences for Future Self. While this is inconsistent in terms of preferences, it is more consistent in terms of explanatory power of the payoffs. We are interested in what guides the agent to action. So, this version of the tradeoff is best suited for the model's purpose. To be clear, this is one of the features of addiction that I aim to highlight in this paper: the kind of irrationality at play in addiction is not describing how one's decision-making fits against her predictions of reward. That is why rational choices are not a prophylactic against it. Addicts suffer from irrational desires, not irrational decisions.

4. Modifying the Payoffs

Perhaps my result appears too narrow in scope because of the way I have set up the payoffs and probabilities for the particular inequality I used. One might ask: How does the inflection point change given different payoffs—that is, different ratios of harm of addiction to benefits of dependence?

Let me introduce a new term K , where K represents the ratio of harm of addiction to the benefit of dependence. Methamphetamines (meth) might have a K -value of 100. Meaning: it is 100 times worse to be addicted to meth than it is beneficial to be dependent on it. Thus, the K -value will act, in the model, as a modifier of the benefit of dependence, which we will keep at 2. If the K -value is close to 1, there is a notable risk: it means that the negative effects of addiction would be as harmful as dependence is helpful (2). A value less than 1 is little risk: it would mean

that the negative effects of addiction are less harmful than the dependence would be helpful (Figure 2 has a K-value of 0.5). A value greater than 1 is high risk: the negative effects of addiction would be greater than the benefits are helpful. Thus, *ceteris paribus*, the payoff of Current Self if she gets addicted to meth would be $-2K$, or -200 . Should Current Self ever engage with meth? Put algebraically, Current Self should engage when

$$p(-2K) + (1 - p)(2) \geq p(0) + (1 - p)(-1)$$

or

$$p \leq 3/(3 + 2K)$$

or

$$p \leq 3/203$$

Thus, for most possible values of p (between 0 and 1), when K is 100, the inequality is false because p would have to be $3/203$ or less. Meaning: it would probably be safe to adopt, as a heuristic, the notion: *just say no to things whose harm from addiction is 100x worse than the benefit of successfully depending on them*. That being said, saying no is still technically not always rational, even here. When p is 1.5% or less, it is rational for both players to engage. It might be challenging to find a substance like this—so dangerous yet so non-addictive. In that case, the heuristic would work. However, that does not eliminate the possibility of rational engagement with highly dangerous substances. Even in this most extreme example, incorporating dependence into the model makes it so *just saying no* is technically not always the optimal strategy.

Most substances are not as risky as meth. As the value of K decreases, the range of p , whereby it would be rational for Current Self to engage, increases. Figure 3 depicts how the inflection point changes given different values of K .

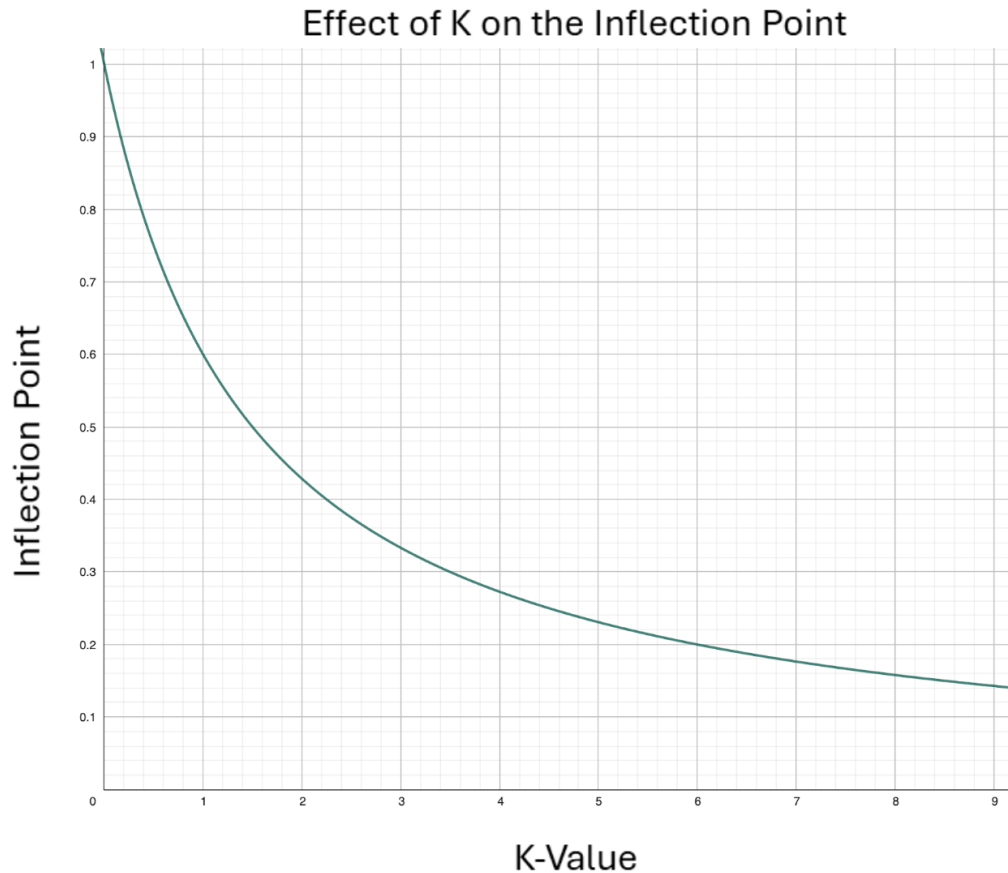


Fig. 3 The effect of K-value on the inflection point. Specifically, this is the curve: $y = 3/(3 + 2x)$, where “y” is the chance that something will end up an addiction, and “x” is the K-value of the substance in question (geogebra.org)

It is of course not obvious exactly what K-value each substance should get for each person.

There is no perfect way of knowing. Maybe, for some people, caffeine gets a K-value of 1, maybe 1.5, maybe 0.5. The point is that, by incorporating dependence into the game theoretic model of addiction, it is clear that people should not *just say no*. Much of the time, actually, we *should say yes*. When it is rational to say yes, but addiction results anyway, it shows us that addiction may result from rational choices.

5. Iterate, Iterate, Iterate

If what I have presented thus far still seems to be missing critical realism, it may be because I have described a one-shot game—as in one decision by Current Self followed by one decision by Future Self, then that game is over. But that doesn't seem right. Wouldn't the game happen a lot? Also, who is this "Future Self" player in real life? Is it Current Self the next time the opportunity to "engage" arises again? Or would that just be Current Self again? To resolve the confusion about who the players are in real life, and how many times the game is played, let us consider an iterated game, where Current Self is you, right now: an individual in the moment of potential engagement. Future Self is every iteration of the game after this given moment. That is, "Future Self" will be modeled as every time stamp after the first one, T1.

Just as before, the rational choice for Current Self will be to engage when the expected payoff from engagement is greater than or equal to the expected payoff from disengagement—in other words, when there is no benefit to switching strategies. However, in this iterated version of the game, we want to incorporate not only the number of iterations one sees herself having, but one's discount of future reward. Then we want a sum of all the predicted benefits of all future instances of the game. So we will need new variables and a new equation to illustrate the effect of discount rate and number of iterations on the value of dependence over time, from the perspective of Current Self.

As before, let us call "p" the probability of addiction. So, "1 - p" is the probability of dependence. Let "a" be a variable that represents the cost of addiction, and "f(x)" be a function representing the value of all future versions of the game put together if it ends up a dependence (this function is where we will incorporate the discount rate and number of iterations). Finally, let us call the cost of disengagement "M." What goes into M? Current Self at T1 has not yet

engaged, so her M does not include a cost of withdrawal. Rather, her M is the cost of missing out on what would have been a successful dependence.

All that put together, assuming Current Self's predictions align with reality, Current Self should rationally engage when:

$$p(a) + (1 - p)(f(x)) \geq M$$

Now I can elaborate on what goes into $f(x)$ —the value of all future versions of the game put together if it ends up a dependence. To craft this function, we need three variables:

1. c : the benefit of dependence.
2. d : the discount rate one has for future reward.
3. x : the number of uses one would ideally experience.

Suppose we had a discount rate of 0.97 and we planned on engaging 10 times. How would that affect the value we place on becoming dependent? The phrase “ $(1 - d)$ ” would tell us what fraction of the original value is removed after discounting: 0.03. If we make it “ $1 - d^{(x+1)}$ ” then we can measure the force of the discount rate over time: about 0.28 in this case. To determine what percent of the original value we use, we calculate: “ $((1 - d^{(x+1)}) / (1 - d))$.” When we multiply that quotient by the benefit of dependence, we are left with c, x periods in the future. So:

$$f(x) = c((1 - d^{(x+1)}) / (1 - d))$$

Again, the rational choice for Current Self will be to engage when the expected payoff from engagement— $p(a) + (1 - p)(f(x))$ —is greater than or equal to the expected payoff from disengagement— M . When it is all written out formally, it reads:

$$p(a) + (1 - p)(c((1 - d^{(x+1)}) / (1 - d))) \geq M$$

To illustrate our example, let us further assume: $a = -31$, $p = 0.5$, $c = 2$, and $M = -10$. In this very specific case, we would find: $-6 \geq -10$. Since the phrase is true, the rollback equilibrium would be for Current Self to engage.

There are a great deal of possible versions of this inequality, depending on the values used for the variables. My point is that incorporating the possibility of dependence into the addiction game makes it plausible that one could rationally engage with a possibly addicting substance, especially if the predicted benefit of dependence is high. We can change the probabilities, change the ratio of possible harm to possible good, make it an iterated game, and it still remains that the possibility of addiction may compete with the possibility of dependence, such that one could rationally engage with a substance that she is aware may lead to addiction.

8. Conclusion

I am not trying to capture a general structure of engagement nor of addiction. Instead, I am showing that addictions can arise through rational decision making. Because I am trying to show that rationality can lead to addiction, the use of this perfectly rational person works as an ideal thought experiment, being charitable to an opponent who may argue that perfect rationality prevents addiction. If it turns out that hardly anyone can even approach such a level of

rationality, then my point is not any less true, even if unrealistic, because, again, I am merely noticing that the pursuit of rationality will not always prevent addiction. Using game theory, I showed when engagement with potentially dependable domains, despite the risk of addiction, is rational.

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